Full Strata Seismic Waveform Inversion With Adaptive Iteration

Wenge Liu, *Member, IEEE*, Qisong Mou, Gulan Zhang, *Member, IEEE*, Wenmao Tu, and Haoze Qin

***Abstract*— In the oil and gas exploration, delineating the tectonic structure of geological targets and predicting physical properties are paramount. As we all know, seismic velocity modeling is crucial for characterizing reservoirs. Enhancing seismic inversion accuracy to obtain quantitative parameters is vital in oil reservoir exploration. Current seismic waveform inversion utilizes the amplitude and timing of seismic records to reconstruct the velocity model, thereby refining the model for seismic migration. This article addresses the convergence and computational challenges in conventional inversion by proposing an adaptive iteration method for step updates. This method outperforms inexact linear searches and empirical formulas by updating the model effectively without additional seismic simulations. In the past, gradient optimization typically requires three seismic modeling in a single iteration, whereas adaptive iteration only requires two, offering practical cost savings. To address stability issues in complex scenarios, we combine parabolic search with adaptive iteration, which is the modified optimization technique, to enhance the convergence rate of objective function. Furthermore, by employing gradient preconditioning based on illumination and multigrid strategy, we achieve full strata velocity modeling from shallow to deep layers. Numerical tests on the theoretical anomaly model and BP model validate the effectiveness and precision of the adaptive iteration, significantly reducing the computational load of seismic inversion and accurately estimating deep strata information.**

***Index Terms*— Adaptive iteration, multiple grid, numerical simulation, seismic waveform, velocity modeling.**

I. Introduction

THE geophysical inversion deduces the Earth's internal structure and physical parameters from diverse observation data. This theory is applied across various fields. Seismic inversion, pivotal in reservoir characterization, becomes critical when oil and gas exploration encounters complex scenarios. The challenge lies in enhancing seismic inversion accuracy to extract quantitative attributes [1]. In deep reservoir exploration, the objective extends beyond mapping structural forms to predicting physical properties and hydrocarbon presence. However, uncertainties in medium velocity impede seismic imaging [2]. Therefore, the estimation of seismic velocity is a major problem in deep reservoir exploration.

Seismic waveform inversion, grounded in least squares optimization, significantly influences geophysical inversion. It employs the L_2 norm of the discrepancy between actual and synthetic seismograms as an objective function, refining the model via local optimization methods [3]. Initially, due to immature theory and limited computing technology, waveform inversion was not widely adopted [4]. Subsequent advancements introduced gradient calculation through the adjoint state method, streamlining the inversion process [5]. Nonlinear challenges in waveform inversion have consistently attracted our attention. Studies reveal data frequency's sensitivity to disturbance scales, underpinning multiscale inversion strategies [6].

Inversion outcomes on seismic data underscore the resolution and initial model dependency [7]. The multiscale strategies, leveraging low-frequency data for large-scale modeling followed by high-frequency data for detailed structure depiction, have proven successful [8]. This approach mitigates inversion nonlinearity, enhancing stability and accuracy. To comprehend crust–mantle interactions, a variable grid-based scale series was devised, enabling inversion across all Earth scales [9]. Frequency-domain seismic data attenuation and objective function design facilitate multiscale inversion [10], [11]. Processors may employ envelope inversion algorithms for noise-resistant frequency component selection or wavenumber component extraction for low wavenumber inversion [12], [13], [14], [15]. Local inversion algorithms reduce high-frequency inversion computations, boosting efficiency [16]. Innovative methods continue to refine inversion accuracy and circumvent periodic hopping [17], [18], [19], [20], [21].

In geophysical inversion, Newtonian and conjugate gradient methods are computationally intensive as they require the inversion of Hessian matrix and determination of conjugate direction. Conversely, gradient algorithms are preferred for their simplicity and widespread application. The iterative step in seismic inversion typically involves parabolic search, inexact linear search, or empirical formulas, all demanding one or more seismic simulations to ascertain the appropriate step, thus prolonging the inversion process [22], [23]. To mitigate computational load, one approach utilizes three-point parabolic fitting with aliasing data, enabling simultaneous multiple shots,

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and reducing dependency on the number of sources [24]. Another method employs the OBFGS algorithm with an adaptive step, which bypasses seismic modeling, although its efficiency hinges on the initial step [25]. Alternatively, synthesizing multiple shots at once to calculate the step size can significantly curtail inversion time, requiring just one extra numerical simulation per iteration [26]. Techniques, such as dynamic small-batch optimization and Anderson iterative acceleration, can enhance the convergence rate of the misfit function [27], [28], [29].

Regularization is pivotal in addressing the ill-posed nature of seismic waveform inversion, with Tikhonov regularization being a common choice. Research indicates that sparse structure constraints benefit stratigraphic continuity and edge delineation [30]. Integrating ground seismic records with well data can enhance seismic inversion resolution and deep imaging, while envelope-based waveform inversion with local angle domain illumination compensation offers additional advantages [31], [32], [33], [34].

To reduce the computational overhead of seismic waveform inversion, we introduce an adaptive iteration method that diverges from traditional inexact linear search, parabolic search, and empirical formulas, determining the step size without extra computations. This method effectively decreases the time required for waveform inversion. Considering the instability for complex models, a modified adaptive method has been developed. To enhance inversion accuracy, we simultaneously implement the multiple grid and source–receiver illumination strategies, achieving comprehensive velocity modeling across all strata.

II. Methodology

A. Seismic Waveform Inversion

Guided by the least squares' optimization theory, we define the objective function as the sum of squared data errors

$$s(\mathbf{v}) = \frac{1}{2}\|\mathbf{d}_{\text{obs}} - \mathbf{d}_{\text{cal}}\|_2^2 \tag{1}$$

where bold characters denote vectors, $\mathbf{d}_{\text{obs}}$ represents the seismic observation record, $\mathbf{d}_{\text{cal}}$ represents the simulated seismic record from the current model, and $\mathbf{v}$ represents the medium velocity.

For large-scale seismic problems such as velocity modeling, a local optimization method typically resolves the equation. The common iteration formula is

$$\mathbf{v}^{k+1} = \mathbf{v}^k + \alpha^k \mathbf{p}\left(\mathbf{v}^k\right) \tag{2}$$

where α^k is the step size and $\mathbf{p}(\mathbf{v}^k)$ is the velocity update direction for the kth iteration.

The inversion's update direction is usually the negative gradient. Employing the adjoint state method to compute the objective function's gradient, for a 2-D seismic issue, yields

$$g(x, z) = \frac{2}{\mathbf{v}^3}\sum_t \frac{\partial^2 \mathbf{u}}{\partial t^2}\hat{\mathbf{u}} \tag{3}$$

where $g(x, z)$ signifies the gradient at the velocity model (x, z), $\mathbf{u}$ is the forward-propagation wavefield, and $\hat{\mathbf{u}}$ is the backpropagated residual wavefield.

Gradient calculation involves cross-correlating the second derivative of the forward-propagation wavefield and backpropagated wavefields. Utilizing a seismic prestack dataset for inversion necessitates two numerical simulations per iteration, simulating the seismic record from the current model and backpropagating wavefield from the residual. With numerous shot gathers, the computational demand escalates significantly.

Model updates typically employ parabolic search, inexact linear search, or empirical formulas to calculate the step size, all requiring seismic simulations to pinpoint the suitable step. The adjoint state method's computational burden stems from two wavefield simulations; in other words, it requires additional computational cost. We suggest an adaptive iterative approach that omits additional seismic modeling, substantially enhancing inversion efficiency.

For stable convergence during model updates, the parabolic search method can determine the update step, with just one seismic modeling, avoiding multiple linear searches. Starting with a global trial step α_{test}, the corrected model is computed, $\mathbf{v}_{\text{test}}^k = \mathbf{v}^k + \alpha_{\text{test}}\mathbf{p}^k$, followed by seismic simulation with the new model $\mathbf{v}_{\text{test}}^k$ to obtain the objective function $s(\mathbf{v}_{\text{test}}^k)$. This linearizes the problem, allowing a second-order Taylor expansion of the objective function (1) near the current model to express

$$s\left(\mathbf{v}_{\text{test}}^k\right) = s\left(\mathbf{v}^k\right) + \alpha_{\text{test}}\left\|\mathbf{p}^k\right\|_2^2 + \frac{\alpha_{\text{test}}^2}{2}\tau + O\left(|\Delta\mathbf{v}|^3\right) \tag{4}$$

where $\Delta\mathbf{v} = \mathbf{v}_{\text{test}}^k - \mathbf{v}^k$ and τ is the parameter related to the residual function's second derivative on model $\mathbf{v}^k$.

Given the objective function $s(\mathbf{v}^k)$ of the kth iteration, the function $s(\mathbf{v}_{\text{test}}^k)$ calculated on the perturbed model, the gradient $\mathbf{p}^k$, and the trial step α_{test}, the parameters τ can be determined

$$\tau = \frac{2 \times \left(s\left(\mathbf{v}_{\text{test}}^k\right) - s\left(\mathbf{v}^k\right) - \alpha_{\text{test}}\left\|\mathbf{p}^k\right\|_2^2\right)}{\alpha_{\text{test}}^2}. \tag{5}$$

If the residual function is quadratic near the model, and $\alpha_{\text{opt}}, \alpha_{\text{test}}$, and τ remain constant, the optimal step α_{opt} is computed using the parabolic search method

$$\alpha_{\text{opt}} = \frac{2\left\|\mathbf{g}^k\right\|_2^2}{\tau}. \tag{6}$$

This process involves setting a trial step, deriving a perturbed model, simulating the seismic wavefield, and optimizing the error function to calculate the optimal iterative step, thus achieving an accurate model update.

B. Adaptive Iterative Update

Assuming that the objective function $s(\mathbf{v}^k)$ is second-order continuously differentiable, the first derivative of the objective function $\nabla s(\mathbf{v}^k)$ can be expressed in the vicinity of $\mathbf{v}^{k+1}$ as follows:

$$\nabla s\left(\mathbf{v}^k\right) = \nabla s\left(\mathbf{v}^{k+1}\right) + \nabla^2 s\left(\mathbf{v}^{k+1}\right)\left(\mathbf{v}^k - \mathbf{v}^{k+1}\right) + O\left(|\Delta\mathbf{v}|^2\right). \tag{7}$$

Defining $\mathbf{y}^k = \nabla s(\mathbf{v}^{k+1}) - \nabla s(\mathbf{v}^k)$ as the difference in the objective function gradient between two adjacent iterations and $\mathbf{m}^k = \mathbf{v}^{k+1} - \mathbf{v}^k$ as the model update, (7) simplifies to

$$\mathbf{y}^k = \nabla^2 s\left(\mathbf{v}^{k+1}\right)\mathbf{m}^k. \tag{8}$$

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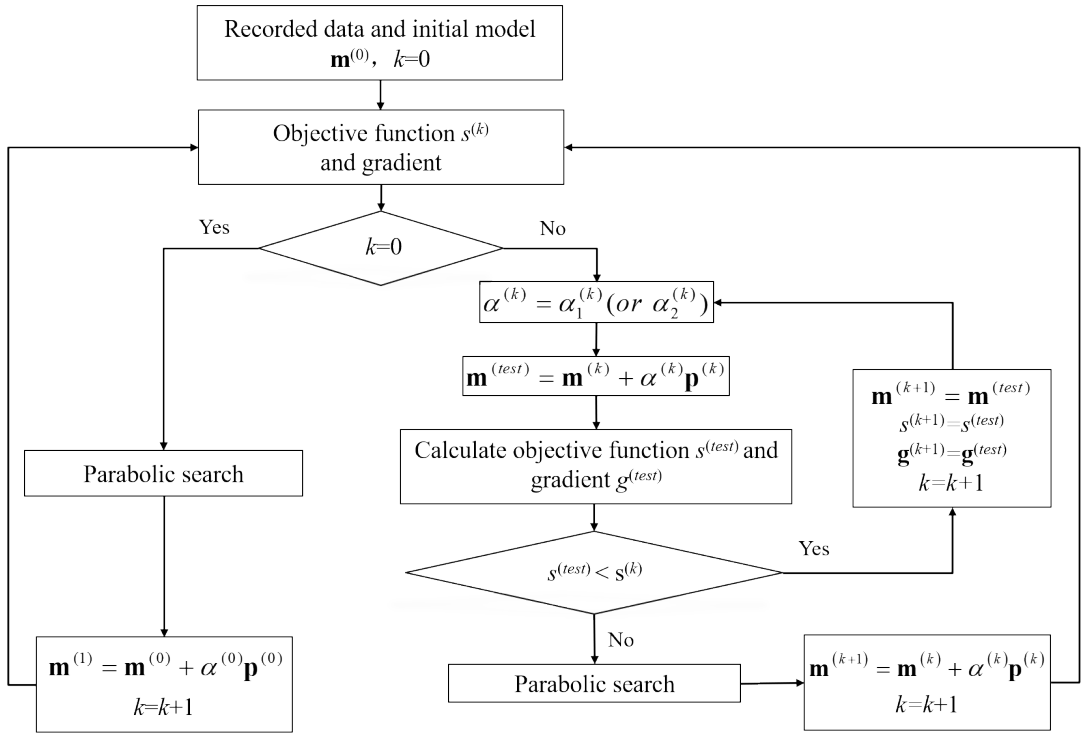

Fig. 1. Process of modified adaptive iteration.

Given that the second derivative of objective function with respect to velocity is the Hessian matrix, the following equation is obtained:

$$\mathbf{y}^k = \mathbf{H}^{k+1}\mathbf{m}^k, \quad \text{or } \left(\mathbf{H}^{k+1}\right)^{-1}\mathbf{y}^k = \mathbf{m}^k. \tag{9}$$

For conventional gradient descent methods, the iterative formula is

$$\mathbf{v}^{k+1} = \mathbf{v}^k - \alpha^k \nabla s\left(\mathbf{v}^k\right). \tag{10}$$

The formula can also be represented in a matrix form

$$\mathbf{v}^{k+1} = \mathbf{v}^k - \mathbf{D}^k \nabla s\left(\mathbf{v}^k\right). \tag{11}$$

In this equation, $\mathbf{D}^k = \alpha^k \mathbf{I}$, which $\mathbf{D}^k$ should possess the properties of a quasi-Newton optimization method, requires altering the solution of optimal step to determine the extreme value of mathematical problem. Consequently, $\mathbf{D}^k$ should be the solution of any equation in the following set:

$$\begin{aligned} &\min_{\alpha^k} \left\|\mathbf{m}^{k-1} - \mathbf{D}^k\mathbf{y}^k\right\|_2^2 \\ &\min_{\alpha^k} \left\|\left(\mathbf{D}^k\right)^{-1}\mathbf{m}^{k-1} - \mathbf{y}^k\right\|_2^2. \end{aligned} \tag{12}$$

Any equation in formula (12) can be utilized to calculate the corresponding iteration step

$$\begin{aligned} \alpha_1^k &= \frac{\left(\mathbf{m}^{k-1}\right)^T\mathbf{y}^{k-1}}{\left(\mathbf{y}^{k-1}\right)^T\mathbf{y}^{k-1}} \\ \alpha_2^k &= \frac{\left(\mathbf{m}^{k-1}\right)^T\mathbf{m}^{k-1}}{\left(\mathbf{m}^{k-1}\right)^T\mathbf{y}^{k-1}}. \end{aligned} \tag{13}$$

According to (13), only two adjacent gradients and model information are needed to calculate the iteration step. There are alternative schemes for this calculation. Theoretical derivation and numerical tests suggest that a smaller iteration step induces a more favorable downward direction in the inversion process, while a larger iteration step produces a sufficient update [35]. This article presents a selection standard for the adaptive step based on theoretical analysis

$$\alpha^k = \begin{cases} \alpha_2^k, & \text{if } \alpha_2^k/\alpha_1^k < 0.5 \\ \alpha_1^k, & \text{otherwise.} \end{cases} \tag{14}$$

C. Modified Adaptive Method

The adaptive step method, as previously mentioned, theoretically accelerates the convergence of seismic inversion. However, in complex situations, it may exhibit instability, such as fluctuations in the objective function or suboptimal inversion of high-frequency components. For specific details, refer to subsequent numerical tests. To address the instability of seismic inversion, we propose a modified adaptive iteration that integrates parabolic search with the adaptive step, as illustrated in Fig. 1.

In practice, the modified adaptive method involves the following steps.

1) In the initial iterative inversion, use parabolic search to determine the step, calculate objective function s^1, and update the velocity model.
2) For iterations where $k > 1$, calculate the step using adaptive iteration and select according to the proposed criteria. Compute objective function s_{test} and gradient g_{test} of the updated model.
3) If we obtain $s_{\text{test}} < s^{k-1}$, then $s^k = s_{\text{test}}$, update the model, and proceed to the next iterative cycle; otherwise, determine the step with parabolic search, update the model, and recycle.

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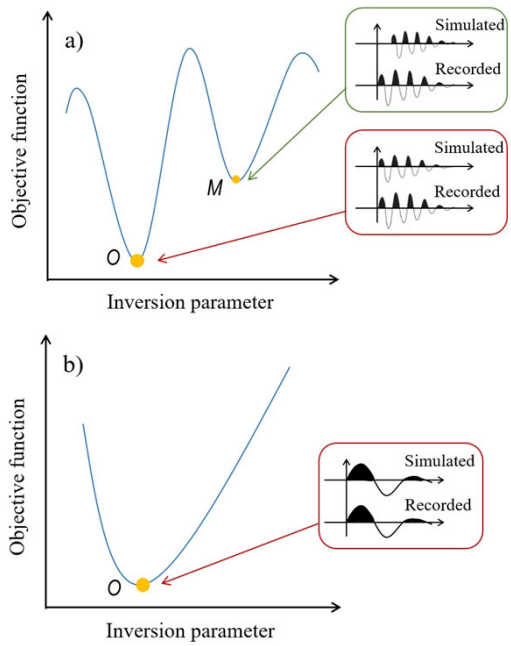

Fig. 2. Objective function optimization under varying conditions. (a) Multi-extremum scenarios prone to periodic jump. (b) Convex function scenarios conducive to unique extrema.

4) Repeat steps 2) and 3) until the convergence condition or the maximum number of iterations is met, and output the final velocity model.

III. Velocity Modeling Strategy

A. Multigrid Method

For the nonlinear problem in seismic inversion, practitioners typically adopt a multiscale strategy, estimating velocity information from low to high frequencies. This approach effectively mitigates the nonlinearity of seismic problem, stabilizing inversion iterations and enhancing result accuracy. Seismic waveform inversion, conducted within the Born approximation framework, primarily incurs calculation errors from scattering. Utilizing low-frequency data in the inversion process limits scattering's impact on the seismic wavefield, thereby reducing calculation errors and diminishing the problem's nonlinearity.

As depicted in Fig. 2(a), high-frequency data in inversion leads to multiple peaks and troughs in the seismic record. Small phase discrepancies between observed and simulated records may cause objective functions to mismatch these features. Multiple extrema in the objective function can result in periodic jumps near M, due to the initial model's uncertainty, trapping the inversion process in a local minimum. Conversely, as shown in Fig. 2(b), low-frequency inversion yields a seismic record with singular peaks or troughs within the time interval, rendering the objective function convex. This correspondence between observation and simulation ensures a unique extremum for the objective function, facilitating the global solution's identification. The comparison of Fig. 2(a) and (b) demonstrates that a multiscale strategy significantly reduces seismic inversion's nonlinearity, thereby improving velocity modeling outcomes.

The multigrid method is an efficient solution for the wave equation, featuring a dynamic mesh size in model discretization ranging from coarse to fine. The crux of multigrid technology is the grid conversion process. Typically, the transition from coarse to fine mesh is termed the continuation operator, while the reverse is known as the restriction operator. The continuation operator's essence is to redivide the model according to a new ratio decreasing spacing and increasing grid count, reflecting the model's refinement.

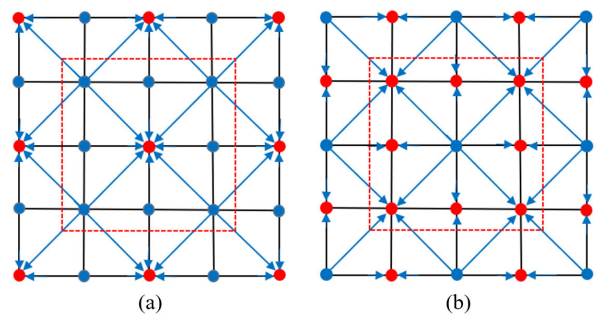

Fig. 3. (a) Restriction operator, with red dots indicating coarse grid vertices and blue dots as fine grid vertices, arrows signifying restriction directions. (b) Continuation operator, with blue dots as coarse grid vertices and red dots as fine grid vertices, arrows denoting continuation directions.

The restriction operator's purpose is to simplify the original model proportionally, thereby increasing grid spacing and reducing grid count, which encapsulates the model's foundational information. In this study, the restriction operator is based on the weighted average method, as illustrated in Fig. 3(a), where red dots denote grid points of large-scale model. Conversely, the continuation operator employs quadratic linear interpolation, detailed in Fig. 3(b). The interpolation is executed in both horizontal and vertical directions, with red dots determined through linear interpolation.

The application of continuation and restriction operators alters the spatial relationship between the shot point and the grid position. For actual seismic data, extraneous data beyond the operator's scope can be judiciously excluded.

In theoretical models, redundant data can be preempted by designing the observation system accordingly. When employing the finite difference method to simulate wavefield, variations in grid spacing may alter the amplitude of wavefield. The relationship between seismic amplitude and grid point quantity is given by

$$C = \frac{A_{\text{ini}}}{A_{\text{con}}} = \frac{N_{\text{con}}}{N_{\text{ini}}} \tag{15}$$

where A_{ini} denotes the original seismic amplitude, A_{con} denotes the seismic amplitude after model scale alteration, N_{ini} denotes the original number of grid points, N_{con} denotes the new grid count, and C denotes the amplitude gain coefficient [36].

Multigrid technology, founded on a multiscale strategy, essentially utilizes large grids for seismic numerical calculations to estimate the model's background information and small grids for computing the high-frequency components. In practice, the inversion frequency of seismic data is established first, followed by the calculation of a spatial interval that satisfies the dispersion condition and a time step determined by stability. The discretization scheme based on the current scale

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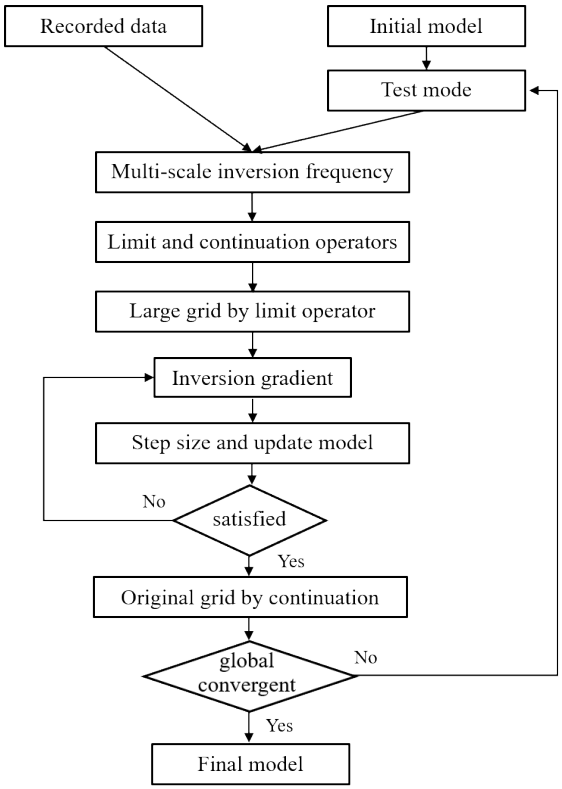

Fig. 4. Flowchart of multigrid implementation.

dictates the resampling interval of original data. The specific implementation steps are as follows (refer to Fig. 4).

1) Define the frequency band division and scope of multi-scale inversion according to velocity modeling requirements.

2) For each frequency band, compute the corresponding extension or restriction operator, and determine the grid size and time interval.

3) Employ the background information of large-scale model and the low-frequency component of seismic data to obtain the inversion results at the current scale.

4) Use the extension operator to revert to the original scale model. If seismic inversion converges, the final model is obtained; otherwise, enter the high-frequency band cycle until the optimal solution is reached.

B. Gradient Preconditioning

In seismic migration, the source or receiver illumination is utilized to suppress artifacts [37]. Seismic waveform inversion may encounter similar challenges. Typically, the source and receiver are positioned on the surface in seismic observation design, resulting in a spatial energy imbalance in the objective function's gradient, complicating model updates from shallow to deep layers. Gradient preconditioning with illumination is applied to enhance the accuracy of seismic inversion. The gradient based on source or receiver illumination is defined as

$$I_s(x,z) = \sum_s \frac{g_s(x,z)}{\sum_t S_s^2(t,x,z)}$$
$$I_r(x,z) = \sum_s \frac{g_s(x,z)}{\sum_t R_s^2(t,x,z)} \tag{16}$$

where I_s is the gradient with single source illumination, I_r is the gradient with single receiver illumination, S_s is the perturbation of forward-propagation wavefield at the position (x, z) and time t, and R_s is the perturbation of receiver backpropagation wavefield at the same position, calculated from synthetic data. The denominator in (16) represents the illumination at the source and receiver. Seismic inversion theory posits that the calculation of Fréchet derivatives is contingent upon both the source forward-propagation and receiver backpropagation wavefields. Consequently, illumination compensation exclusively at the source or receiver is insufficient.

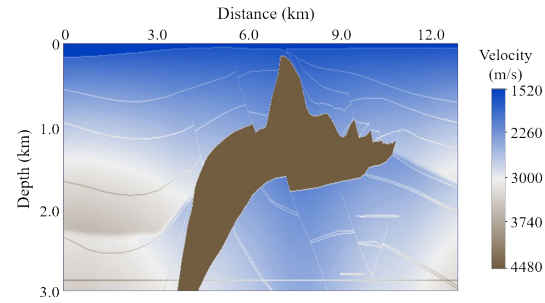

Fig. 5. SEG/EAGE 2-D salt model.

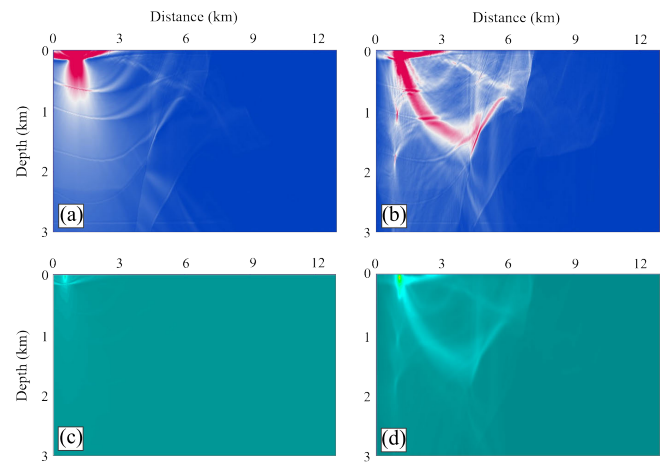

Fig. 6. Illuminations of salt model. (a) Source forward-propagation wavefield illumination. (b) Coupled source–receiver illumination. (c) and (d) Normalized energy profiles of (a) and (b).

A proposed method amalgamates illumination data from both the source and the receiver to enhance reverse time migration (RTM) and achieves superior imaging outcomes [38]. The bidirectional illumination in inversion is defined as

$$\tilde{g}(x,z) = \sqrt{I_s(x,z) I_r(x,z)} \tag{17}$$

where $\tilde{g}(x, z)$ is the gradient after preconditioning with source and receiver illumination.

The SEG/EAGE salt model (referenced in Fig. 5) provides an example for evaluating the efficacy of traditional compensation alongside coupled source–receiver illumination. Within the model's central region, a high-velocity rock formation extends extensively, obstructing the downward and lateral transmission of seismic waves. The finite difference method is employed for seismic modeling, with both vertical and horizontal mesh spacing set at 10 m. The source is positioned 1000 m horizontally from the origin and at a depth of 20 m. Receivers are arrayed across the model's summit, spaced 10 m

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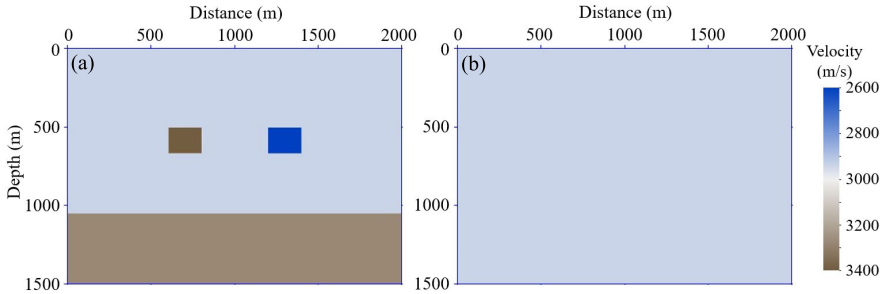

Fig. 7. Abnormal body model. (a) Actual model. (b) Initial model.

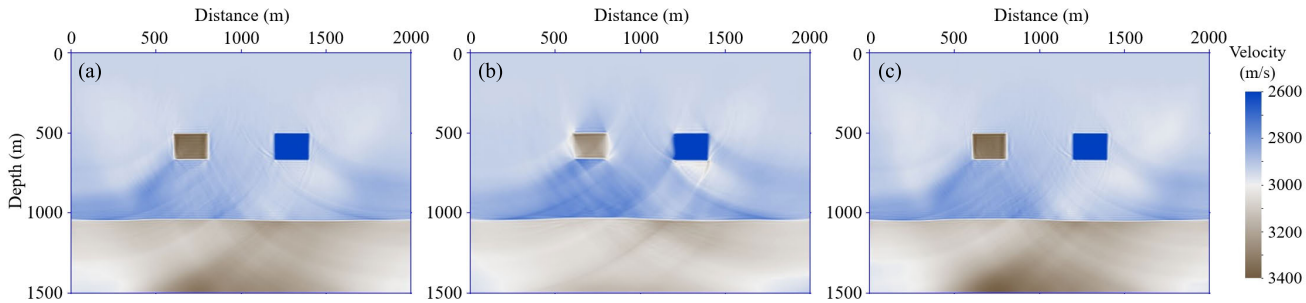

Fig. 8. Velocity modeling using various methods. (a) Parabolic search. (b) Adaptive iteration. (c) Modified adaptive.

apart. An illumination analysis is conducted to assess the energy distribution, which is depicted in Fig. 6.

Seismic illumination serves to equalize energy distribution throughout seismic wave propagation. The spatial analysis of wavefield reveals the following.

1) Source illumination primarily reflects the energy distribution of the transmitted wavefield, with intensity concentrated near the source and diminishing outward. Given its insensitivity to pronounced reflective interfaces within the model, source illumination predominantly affects proximal areas, exerting minimal influence on imaging deeper geological layers, as illustrated in Fig. 6(a).
2) Coupled source–receiver illumination captures the energy distribution of both transmitted and reflected wavefields. It accounts for the influence of both the source and receivers on energy distribution. In theory, seismic inversion that incorporates bi-illumination should concurrently enhance imaging near the source and within deeper strata.

IV. Numerical Examples

A. Abnormal Body Model

This section demonstrates the efficacy of adaptive iterative method using an anomalous body model. The real velocity model, depicted in Fig. 7(a), consists of a two-layer structure with velocities of 3000 and 3300 m/s, respectively. Within the shallow layer's midpoint, two anomalies signify high- and low-velocity regions, with velocities of 3400 and 2600 m/s, respectively. The initial model for inversion is displayed in Fig. 7(b). The models are discretized with 400 transverse and 300 longitudinal grid points, each measuring 5 × 5 m. Seismic observations involve arranging excitation and receiver points on the surface, with a total of ten shots discharged.

The seismic survey employs a shot spacing of 180 m, with the first shot's abscissa at 180 m, and a full arrangement of receivers. The finite difference method, utilized for seismic modeling, operates at an eighth order in space and a second order in time. A Ricker wavelet is adopted as the source with a main frequency of 40 Hz, and a time sampling interval is 0.5 ms. Throughout the inversion process, velocity constraints range from 2400 to 3600 m/s. Seismic waveform inversion incorporates parabolic search, adaptive iteration, and a modified adaptive method for step size calculation.

The experimental setup includes an Intel[1] Xeon[1] Silver Class 4110 CPU, operating at 2.1 GHz (3.0-GHz Turbo), with eight CPUs, 9.6-GT/s 2UPI, and 32-GB RAM. Six threads facilitate the actual computation. To mitigate the inversion nonlinearity, a multiscale strategy segments data frequency into 2, 5, 10, 20, and 40 Hz, progressing sequentially from low to high frequency.

Fig. 8 presents the inversion results. A comparative analysis of velocity outcomes indicates that the modified adaptive method outperforms the other two approaches. This superiority is evident in the delineation of the anomalous body boundary, where the velocity aligns more closely with the actual model and background disturbances are minimized.

A transverse velocity profile, extracted from a depth of 600 m as depicted in Fig. 9, facilitates the analysis of inversion error. Examination of two anomalous bodies' outlines reveals superior results for the low-velocity anomaly. Among the methods compared, parabolic search, adaptive iteration, and modified adaptive iteration, the latter yields the most accurate boundary delineation for the high-velocity anomaly.

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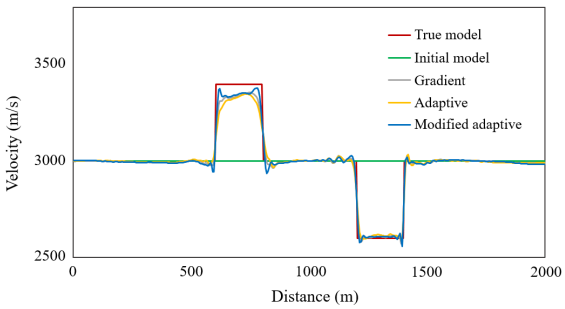

Fig. 9. Comparison of velocity curves.

TABLE I
TIME FOR DIFFERENT OPTIMIZED INVERSIONS

Inversion frequency	Parabolic Search (s)	Adaptive iteration (s)	Modified adaptive (s)
2Hz	1221 (7)	896 (7)	894 (7)
5Hz	1234 (21)	897 (14)	891 (14)
10Hz	1234 (14)	894 (22)	896 (22)
20Hz	1231 (49)	897 (19)	1036 (31)
40Hz	1228 (50)	890 (19)	1038 (38)

Note: The table values represent the duration of a single iteration at each scale, with the number of iterations indicated in parentheses.

Inversion outcomes indicate that modified adaptive iteration is the most effective, although adaptive iteration occasionally matches the parabolic search's performance. The methods' efficacy is evaluated based on the convergence of objective function. Fig. 10 illustrates the objective function's variation with iterations across different scales, highlighting the adaptive iteration's instability, particularly its convergence challenges at higher frequencies.

At a data frequency of 40 Hz, the objective function's fluctuation intensifies, resulting in unsatisfactory inversion velocities. The modified adaptive iteration balances the strengths of parabolic search and adaptive approach. During the background information inversion phase, at data frequencies of 5 and 10 Hz, the objective function decreases rapidly. Even at 20 and 40 Hz, this method maintains the objective function's stability, thereby enhancing inversion accuracy to a notable degree.

The primary objective of adaptive iteration within this study is to diminish the computational time required for seismic waveform inversion. Table I delineates the time expenditure associated with anomalous bodies model. A single iteration utilizing parabolic search consumes approximately 1230 s, whereas adaptive iteration requires only 895 s, marking a reduction of roughly 28%.

The duration for modified adaptive method is extended, and its application in high-frequency inversion, where parabolic search aids in enhancing the convergence of objective function, leads to an increased computational time. Nevertheless, the time is curtailed by 16%. By amalgamating the duration of a single iteration with the total number of iterations, the cumulative time for various methods is ascertained. The parabolic search accounts for 173 456 s, adaptive iteration for 72 451 s, and modified adaptive method for 110 004 s. In comparison, adaptive iteration achieves a time reduction of approximately 59%, while the modified adaptive method records a decrease of 37%.

B. BP Velocity Model

The BP model of SEG/EAGE is frequently employed to test velocity modeling, and for this purpose, we utilize it to assess the effectiveness of strategy such as multigrid and gradient preconditioning in full strata velocity modeling. The width of BP model is 67 425 m and the depth is 11 937.5 m, with its structure illustrated in Fig. 11. It has a velocity range of 1429–4790 m/s and a spatial interval of 12.5 × 6.25 m and comprises 53 951 911 grids.

Due to the scale of BP model, implementing inversion on standard computing devices is impractical; therefore, we have simplified the model. We restricted the model's depth to 10 000 m, considering the limited depth for oil and gas exploration. For the transverse span of BP model, we divided it into three sections, designated as models A, B, and C, and the boundary positions are marked by orange lines in Fig. 11. During the actual inversion process, each segment's lateral boundary is extended by 5 km, to mitigate artificial boundary effects. For seismic observations, source and receiver points are positioned on the surface, with the initial shot at $x =$ 700 m, and intervals of 1500 m. The excitation source is a Ricker wavelet with a dominant frequency of 10 Hz. The grid interval is set at 25 m, the seismic simulation time is set at 16 s, and the interval is set at 2 ms. The synthetic dataset is generated using an eighth-order spatial and second-order temporal finite difference method, with a P-wave velocity and fixed density. The initial model, derived from the BP model with Gaussian smoothing and a radius of 1250 m, is refined through ten iterative cycles.

Now, model B serves as a representative example to illustrate the inversion process. Seismic waveform inversion is executed at 0.5, 1, 2, and 5 Hz, employing with multigrid technique. For the 5-Hz inversion, the spatial sampling interval is maintained at 25 m. At the inversion frequency of 1 and 2 Hz, the spatial interval is augmented to 50 m, and the time interval is still 2 ms, resulting in a grid reduction to one-fourth. For the 0.5-Hz frequency, the spatial grid is increased to 100 m and the temporal interval is 4 ms, significantly diminishing the number of grids and seismic data, thereby enhancing efficiency.

The number of iterations for each scale is set at 100, with the process terminating once the relative change in the objective function falls below 1%. Fig. 12 depicts the L_2 norm residual's relationship with iterations, demonstrating objective function's convergence at various scales. The data within the figure are normalized by the objective function value of first inversion. This visualization indicates that, during the inversion of background information, there is a swift decrease in the objective function, followed by varying rates of decline in subsequent stages. On the whole, it is initially rapid, then decelerating, and ultimately approaching stability.

The efficiency of seismic inversion and its acceleration are quantitatively analyzed. Utilizing the original grid and

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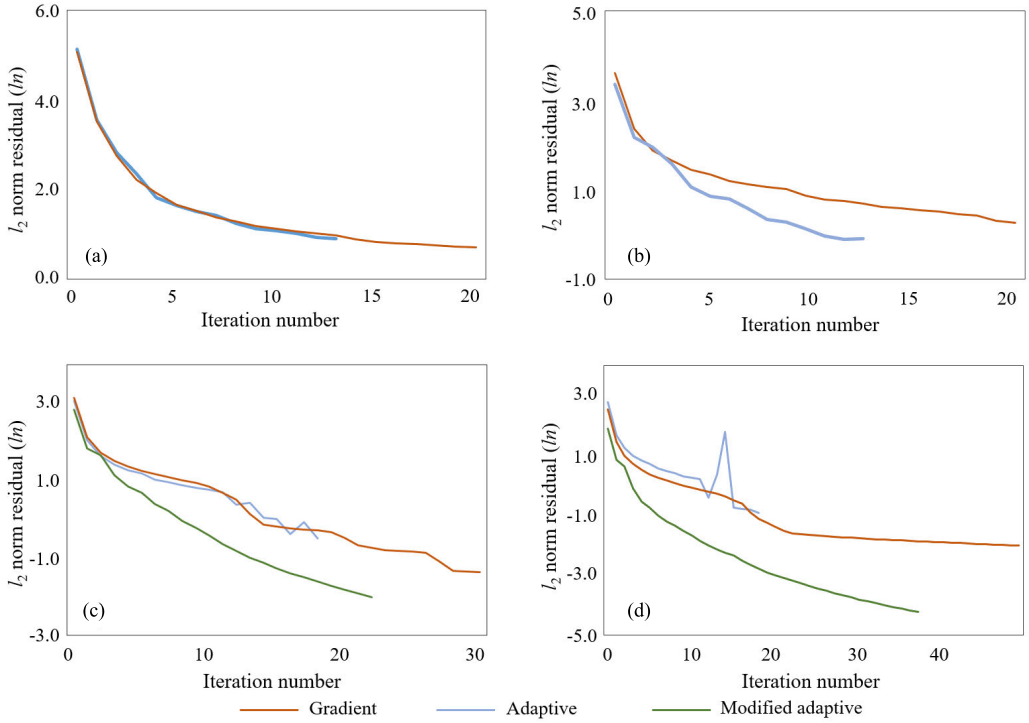

Fig. 10. Objective function's relationship with iterations. (a) 5 Hz. (b) 10 Hz. (c) 20 Hz. (d) 40 Hz.

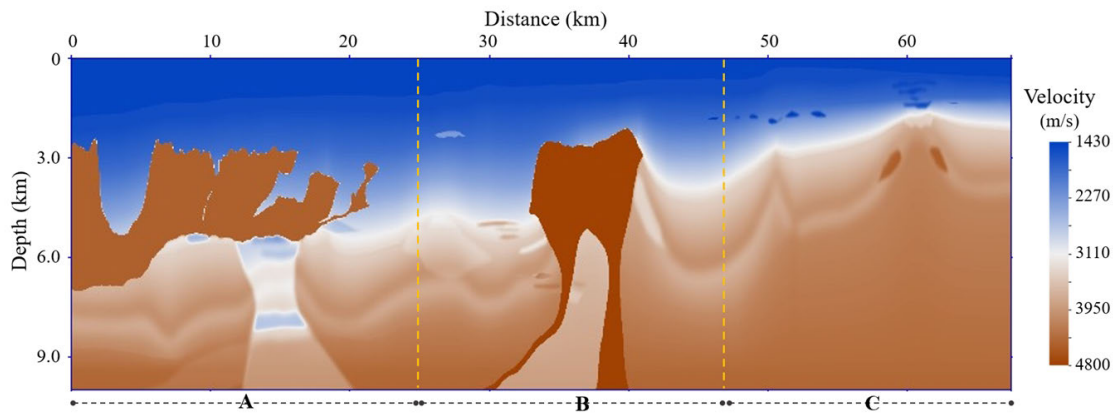

Fig. 11. BP velocity model.

establishing an acceleration ratio of 1, a single iteration takes 10 231 s. Reducing the spatial grid size to one-fourth, the iteration time is 1764 s with an acceleration ratio of 5.8; further reducing the grid to one-sixteenth lowers the time to 314 s and achieves an acceleration ratio of 18.85. Therefore, the multigrid technique significantly reduces the time spent on seismic inversion.

The gradient field is computed from the backpropagation of residual wavefield and the second derivative of source wavefield forward propagation. Given the BP model's inversion depth of 10 000 m, seismic wave amplitude varies with spherical diffusion during propagation and is also influenced by the source and receiver positions. This results in distinct gradient values for each grid, which can adversely affect the deep layer's representation upon model updating.

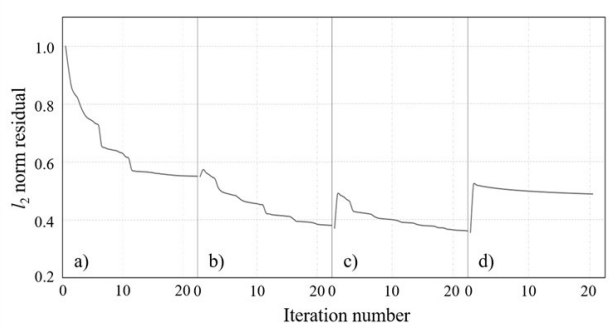

Fig. 12. Objective function's relationship with iterations at different scales. (a) 0.5 Hz. (b) 1 Hz. (c) 2 Hz. (d) 5 Hz. Note: values of objective function in these figures are normalized.

To enhance the inversion accuracy of deep strata, the gradient of objective function is regularized using illumination

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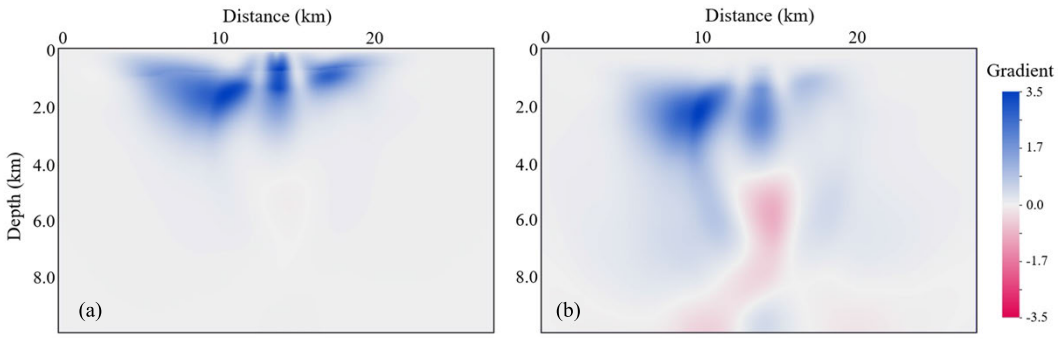

Fig. 13. Gradient preconditioning. (a) Original data. (b) Preconditioning based on source–receiver illumination.

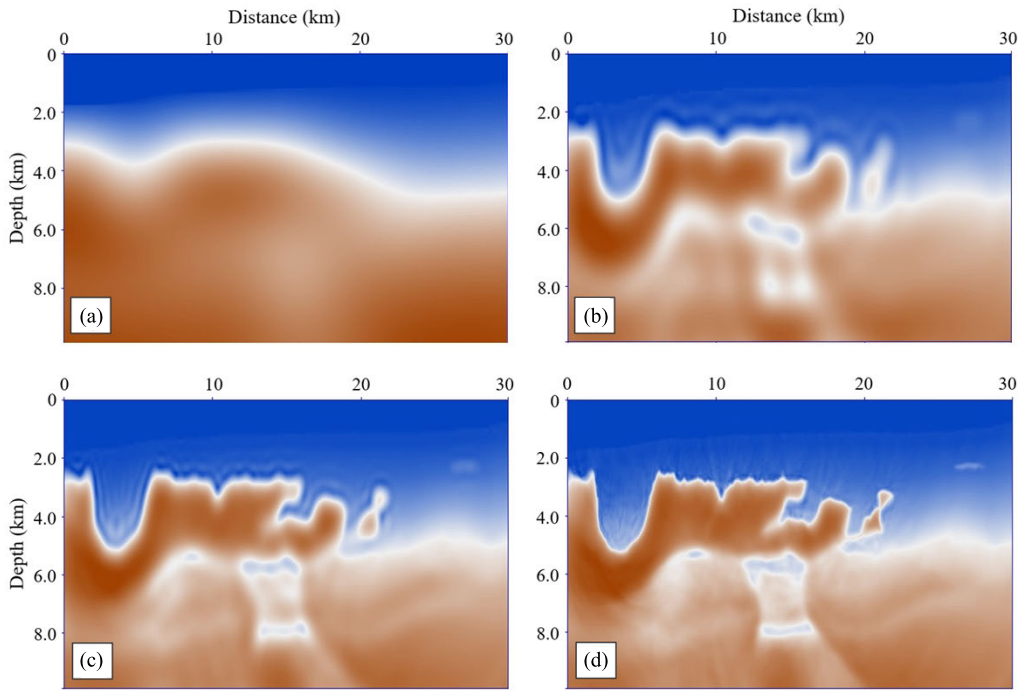

Fig. 14. Inversion results for model A. (a) Initial velocity model. (b) 1 Hz. (c) 2 Hz. (d) 5 Hz.

technology, which is based on seismic record simulations, and ensuring a spatially balanced gradient field. The outcomes of regularization are displayed in Fig. 13. The inversion results for each submodel, as shown in Figs. 14–16, maintain color scales consistent with Fig. 11.

Subsequent analysis compares velocity modeling at different scales. In low-frequency inversion, the impact of the model's large-scale structure is pronounced; at frequencies of 1–2 Hz, the boundaries of salt mounds in models A and B are discernible, and finer structures in shallow regions become increasingly apparent; at a frequency of 5 Hz, only the structural boundaries exhibit alterations.

In summary, velocity modeling of large-scale salt mounds within BP model has yielded satisfactory outcomes. However, the delineation of steep boundaries at the depths remains imprecise, and subsalt structures are not distinctly resolved. In contrast, model C, which lacks salt mounds and complex structures in the study area, exhibits no significant issues in seismic inversion across all depths.

Upon comparing the inversion of model B with actual model, it is observed that the structure of BP model is essentially reconstructed. The shallow regions exhibit the most accurate imaging, and minor high-velocity structures and low-velocity anomalies are discernible. Despite the application of preconditioning to the gradient in the inversion process, inaccuracies persist in the deeper regions, particularly at sloping velocity interfaces. However, the fundamental pattern remains discernible. The inversion process across various frequencies substantiates the efficacy of the multiscale strategy.

A horizontal velocity profile extracted from model B at the depth of 5 km, as illustrated in Fig. 17, allows for a

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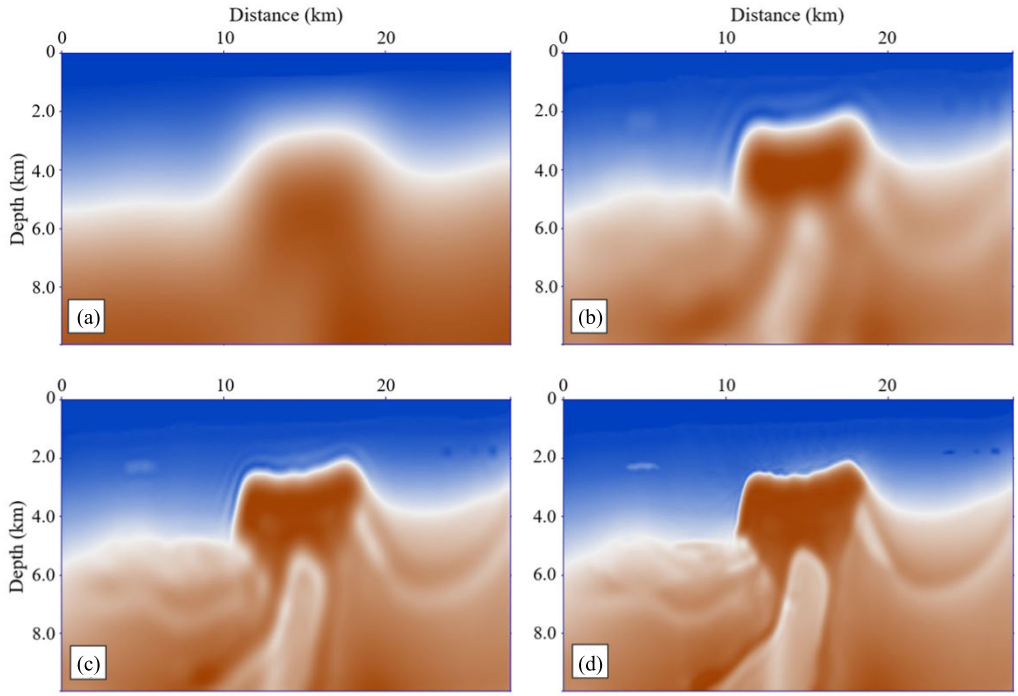

Fig. 15. Inversion results for model B. (a) Initial velocity model. (b) 1 Hz. (c) 2 Hz. (d) 5 Hz.

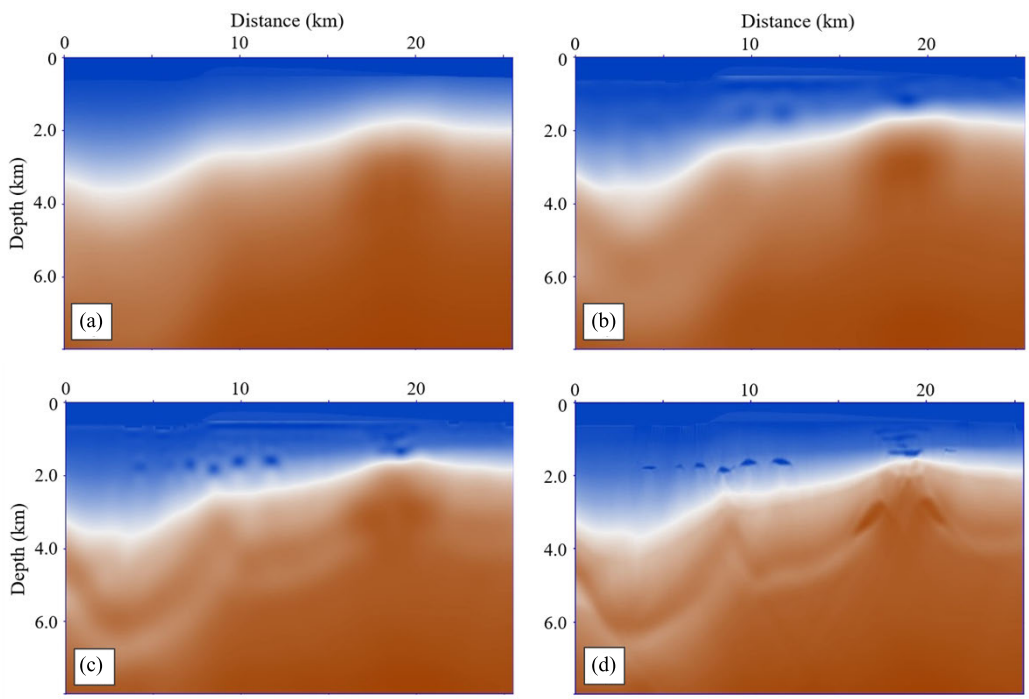

Fig. 16. Inversion results for model C. (a) Initial velocity model. (b) 1 Hz. (c) 2 Hz. (d) 5 Hz.

comparative analysis. The comparison reveals that the velocity discrepancy between the initial and actual models is approximately 2 km/s. Notably, this error is significantly diminished following the inversion process.

Velocity modeling on the left side of model B aligns well, indicating a successful inversion. Conversely, on the right side of the salt, notable inversion errors are present. These errors, particularly significant at the right extremity, are attributed to the lack of wavefield information at the boundary.

Vertical velocity profiles extracted at 10, 15, and 20 km along the distance axis, as depicted in Fig. 18, suggest that the inversion result is superior in shallower regions compared to deeper ones. Moreover, areas with simpler geological structures yield better inversion results than those with more

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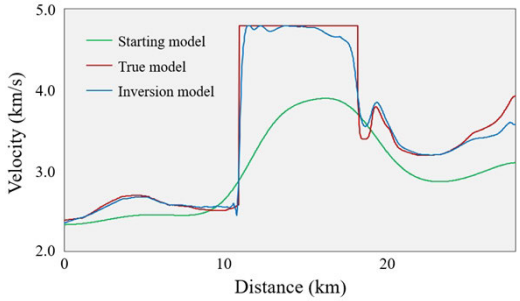

Fig. 17. Horizontal velocity profiles for model B ($z = 5$ km).

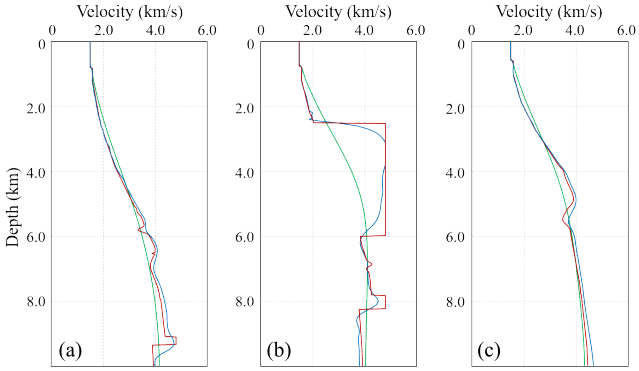

Fig. 18. Vertical velocity profiles for model B. (a) $x = 10$ km. (b) $x = 15$ km. (c) $x = 20$ km.

complexity. While deep velocity is not entirely recoverable, the inversion still effectively captures the essence of geological forms, essentially restoring the construction's shape.

V. Conclusion

This study focuses on full strata nonlinear velocity modeling for complex geological structures. We examined the convergence of seismic waveform inversion and the sensitivity of deep velocity data. An optimization approach for iterative steps is proposed, incorporating strategies such as the multigrid method and gradient preconditioning. The efficacy and practicality of velocity modeling techniques are demonstrated through numerical tests on the anomaly body model and the BP model, yielding the following insights.

1) To diminish the computational demands of seismic inversion, an adaptive method for step determination is introduced, which does not necessitate additional wavefield numerical simulations. While conventional seismic waveform inversion requires three seismic modeling in each iteration, the adaptive iteration method theoretically reduces computation time by one-third. Numerical tests validate the efficiency of this approach. Moreover, to address stability issues encountered during the inversion of high-frequency data, the modified adaptive method proves to be a robust solution for maintaining the stability of seismic inversion.
2) The application of multigrid technology, grounded in a multiscale strategy, significantly mitigates the nonlinearity and computational load of seismic waveform inversion. The spatial grid size and temporal intervals are adaptable across different scales. Employing larger spatial grids and time intervals for low-frequency velocity inversion markedly reduces computational and data storage demands, thereby enhancing the efficiency of seismic waveform inversion.
3) To enhance the precision of deep velocity modeling, the gradient of objective function must be regularized using source–receiver illumination techniques. By integrating the adaptive iteration with a multigrid strategy, velocity modeling of intricate geological bodies can be achieved. These methodologies are substantiated by numerical experiments using the BP model.

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